ORIGINAL ARTICLE

Pharmacist-led home blood pressure telemonitoring after an emergency visit for severe hypertension: a pilot randomised trial

Lucía Ortega Ramos[1], Daniel Achterberg[2], Marta Quiroga Gil[1], Samuel Okafor[3]

[1] Department of Internal Medicine, Hospital General de Almenara
[2] Community Pharmacy Network, Almenara Health Area
[3] Primary Care Research Unit, Almenara Health Area

Illustrative article written for the Postext Cookbook: the trial, its authors and its data are invented. The works cited are real.

ABSTRACT

Background Patients discharged from the emergency department after a visit for severe hypertension often wait weeks for a review, and many return with the same readings.

Methods In this pilot trial, 120 adults seen for a systolic blood pressure of 180 mm Hg or more were randomised to home telemonitoring with medication titration by a community pharmacist, or to usual care. The primary outcome was office systolic pressure at 12 weeks.

Results At 12 weeks, mean systolic pressure was 136.2 mm Hg with telemonitoring and 145.9 mm Hg with usual care (adjusted difference −9.4 mm Hg; 95% CI −13.8 to −5.0). Control below 140/90 mm Hg was reached by 63% and 37% of patients.

Conclusions Telemonitoring led by pharmacists lowered blood pressure in the weeks after an emergency visit. A larger trial with cardiovascular outcomes is justified.

Keywords hypertension; telemonitoring; pharmacists; emergency department; self-monitoring

Introduction

High blood pressure is the leading modifiable cause of cardiovascular death. In 2019 about 1.28 billion adults aged 30 to 79 years had hypertension, and fewer than a quarter of them had it under control.[1] Each reduction of 10 mm Hg in systolic pressure lowers the risk of major cardiovascular events by about a fifth,[2] and a target below 120 mm Hg prevented more events than one below 140 mm Hg in patients at high risk.[3] European guidelines ask for control within three months of starting treatment,[4] and American ones for a review every month until it is reached.[5]

Patients who come to the emergency department with severe hypertension and no acute organ damage are a group the guidelines say little about. Most are sent home with a prescription and a referral to their general practitioner, and the first review may be weeks away. Self-monitoring of blood pressure lowers it when it is combined with a plan for acting on the readings,[6–8] and Margolis et al.[9] showed that pharmacists who adjusted treatment from transmitted home readings improved control for a year in primary care. Whether the same model works in the weeks after an emergency visit has not been tested.

We designed a pilot trial to estimate the effect of pharmacist-led telemonitoring on blood pressure at 12 weeks in this group, and to measure recruitment, retention and safety for a larger trial.

KEY POINTS

What is already known Self-monitoring lowers blood pressure when someone acts on the readings, and pharmacists who manage telemonitored readings improve control in primary care.

What this study adds After an emergency visit for severe hypertension, telemonitoring with titration by a community pharmacist lowered office systolic pressure by 9.4 mm Hg more than usual care at 12 weeks.

Methods

Design and participants

This was a parallel-group, open-label pilot trial with 1:1 allocation, reported according to the CONSORT statement.[10] We enrolled adults aged 18 to 80 years discharged from the emergency department of a district general hospital after a visit with a systolic pressure of 180 mm Hg or more, with no acute organ damage, and living within reach of a participating community pharmacy. Patients with secondary hypertension, an estimated glomerular filtration rate below 30 mL/min/1.73 m², pregnancy or a life expectancy under one year were excluded.

Interventions

Each patient in the telemonitoring group received a validated monitor for the upper arm, which sent every reading to a secure server. They measured twice in the morning and twice in the evening on three days a week. A community pharmacist reviewed the readings weekly and changed

treatment by a written protocol agreed with the general practitioners of the area, aiming at a home mean below 135/85 mm Hg. Patients in the usual-care group were referred to their general practitioner, as is routine.

Outcomes and analysis

The primary outcome was the mean of the second and third of three office systolic readings at 12 weeks, taken by a nurse who did not know the allocation. Secondary outcomes were control below 140/90 mm Hg, return visits to the emergency department and adverse events. Data were collected in REDCap.[11] The difference between groups was estimated by linear regression adjusted for baseline systolic pressure. With 60 patients per group, the trial had 80% power to detect a difference of 8 mm Hg, assuming a standard deviation of 15 mm Hg and 10% loss to follow-up.

Results

Between January and June, 163 patients were screened and 120 were randomised; 57 in the telemonitoring group and 57 in the usual-care group attended the 12-week visit. The groups were similar at baseline (Table 1). Patients in the telemonitoring group sent a median of 31 sets of readings, and the pharmacists made a median of two changes to treatment per patient.

Home systolic pressure in the telemonitoring group fell most in the first four weeks, and by week 10 it was close to the target (Figure 1). At 12 weeks, the mean office systolic pressure was 136.2 mm Hg with telemonitoring and 145.9 mm Hg with usual care, an adjusted difference of −9.4 mm Hg (95% CI −13.8 to −5.0). Control below 140/90 mm Hg was reached by 36 of 57 patients (63%) with telemonitoring and by 21 of 57 (37%) with usual care.

Three patients in the telemonitoring group and nine in the usual-care group returned to the emergency department with raised blood pressure. Two patients in the telemonitoring group reported dizziness on standing after a dose increase, and both recovered after the dose was reduced. No serious adverse events were related to the intervention.

Table 1. Characteristics of the patients at baseline.

Characteristic	Telemonitoring (n = 60)	Usual care (n = 60)
Age, years	58.7 (11.2)	60.1 (10.4)
Women	27 (45)	29 (48)
Systolic BP at the emergency visit	192.4 (10.8)	190.9 (9.6)
Systolic BP at randomisation	161.3 (13.5)	160.2 (12.9)
Diastolic BP at randomisation	94.8 (10.1)	93.6 (9.7)
Diabetes	14 (23)	12 (20)
Chronic kidney disease	6 (10)	8 (13)
Antihypertensive drugs, number	1.4 (1.0)	1.5 (1.1)
No treatment before the visit	19 (32)	17 (28)
Home monitor before the trial	11 (18)	13 (22)

Values are mean (SD) or number (%). BP, blood pressure, in mm Hg.

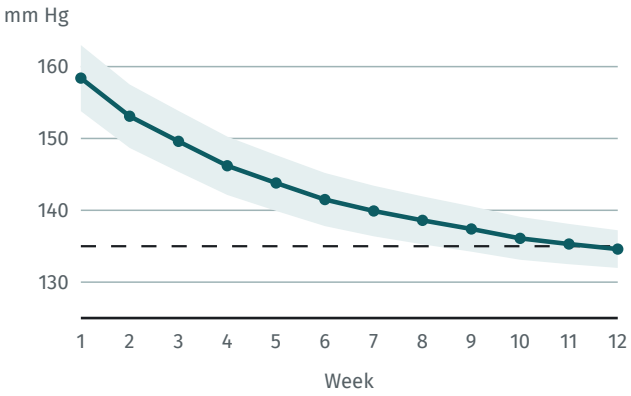


Figure 1. Mean home systolic pressure by week in the telemonitoring group, with its 95% confidence interval. The dashed line is the home target of 135 mm Hg.

Discussion

In this pilot trial, telemonitoring with titration by a community pharmacist lowered office systolic pressure by about 9 mm Hg more than usual care in the 12 weeks after an emergency visit for severe hypertension. The size of the effect is close to the 10 mm Hg that meta-analysis associates with a fifth fewer cardiovascular events,[2] and it was reached within the three months that European guidelines allow.[4]

The result agrees with trials of self-monitoring in primary care, where the benefit came from acting on the readings rather than from taking them.[6,9] Pharmacists were well placed to act. They saw the readings every week, they knew the patients from collecting their prescriptions, and the protocol let them change treatment without waiting for an appointment, as the pharmacists did in the trial by Margolis et al.[9] in Minnesota.

The trial has limitations. It was small and open-label, it was run in one health area, and 12 weeks is too short to show an effect on events. The office readings were taken by a nurse blind to the allocation, which limits the bias that an open design invites. Fewer return visits to the emergency department in the telemonitoring group are encouraging, but the numbers are too small for a conclusion.

A multicentre trial powered for return visits and cardiovascular events is the next step. The rates of recruitment and retention seen here, three in four patients screened and 95% retained, suggest that it is feasible.

Article information

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stitutions and results are invented for this example; the references are to real published works.

References

1. NCD Risk Factor Collaboration (NCD-RisC). Worldwide trends in hypertension prevalence and progress in treatment and control from 1990 to 2019: a pooled analysis of 1201 population-representative studies with 104 million participants. Lancet 2021;398(10304):957–80. https://doi.org/10.1016/S0140-6736(21)01330-1.
2. Ettehad D, Emdin CA, Kiran A, Anderson SG, Callender T, Emberson J, et al. Blood pressure lowering for prevention of cardiovascular disease and death: a systematic review and meta-analysis. Lancet 2016;387(10022):957–67. https://doi.org/10.1016/S0140-6736(15)01225-8.
3. SPRINT Research Group. A randomized trial of intensive versus standard blood-pressure control. N Engl J Med 2015;373(22):2103–16. https://doi.org/10.1056/NEJMoa1511939.
4. Williams B, Mancia G, Spiering W, Agabiti Rosei E, Azizi M, Burnier M, et al. 2018 ESC/ESH guidelines for the management of arterial hypertension. Eur Heart J 2018;39(33):3021–104. https://doi.org/10.1093/eurheartj/ehy339.
5. Whelton PK, Carey RM, Aronow WS, Casey DE, Collins KJ, Dennison Himmelfarb C, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults. Hypertension 2018;71(6):e13–115. https://doi.org/10.1161/HYP.0000000000000065.
6. Tucker KL, Sheppard JP, Stevens R, Bosworth HB, Bove A, Bray EP, et al. Self-monitoring of blood pressure in hypertension: a systematic review and individual patient data meta-analysis. PLoS Med 2017;14(9):e1002389. https://doi.org/10.1371/journal.pmed.1002389.
7. McManus RJ, Mant J, Haque MS, Bray EP, Bryan S, Greenfield SM, et al. Effect of self-monitoring and medication self-titration on systolic blood pressure in hypertensive patients at high risk of cardiovascular disease: the TASMIN-SR randomized clinical trial. JAMA 2014;312(8):799–808. https://doi.org/10.1001/jama.2014.10057.
8. McManus RJ, Mant J, Franssen M, Nickless A, Schwartz C, Hodgkinson J, et al. Efficacy of self-monitored blood pressure, with or without telemonitoring, for titration of antihypertensive medication (TASMINH4): an unmasked randomised controlled trial. Lancet 2018;391(10124):949–59. https://doi.org/10.1016/S0140-6736(18)30309-X.
9. Margolis KL, Asche SE, Bergdall AR, Dehmer SP, Groen SE, Kadrmas HM, et al. Effect of home blood pressure telemonitoring and pharmacist management on blood pressure control: a cluster randomized clinical trial. JAMA 2013;310(1):46–56. https://doi.org/10.1001/jama.2013.6549.
10. Schulz KF, Altman DG, Moher D, CONSORT Group. CONSORT 2010 statement: updated guidelines for reporting parallel group randomised trials. BMJ 2010;340:c332. https://doi.org/10.1136/bmj.c332.
11. Harris PA, Taylor R, Thielke R, Payne J, Gonzalez N, Conde JG. Research electronic data capture (REDCap): a metadata-driven methodology and workflow process for providing translational research informatics support. J Biomed Inform 2009;42(2):377–81. https://doi.org/10.1016/j.jbi.2008.08.010.